Kunjal Purohit

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Education

University of California, Santa Cruz — Santa Cruz, CA
Bachelor of Science, Double Major in Computer Science and Computer Engineering — *Expected June 2028*

- **GPA:** 3.95/4.0
- **Relevant Coursework:** Data Structures & Algorithms, Computer Systems & C, Assembly Language, Discrete Mathematics, Probability & Statistics, Linear Algebra, Multivariable Calculus, Differential Equations

Skills

Languages: Python, TypeScript, JavaScript, C/C++, Java, Swift, SQL (PostgreSQL), HTML/CSS, Assembly
Software Engineering: Object-Oriented Programming, System Design, Testing, Data Pipelines, CI/CD
Frameworks & Libraries: Next.js, React, Node.js, FastAPI, Capacitor, Playwright, Tailwind CSS, OpenCV
Tools & Platforms: Supabase, Neo4j, Clerk, Plaid, Vercel, Codemagic, Unix/Linux
Embedded Systems: STM32, CAN Bus, OneWire, Sensor Integration, Signal Filtering
AI: Coding Agents, LLM Integration, Prompt Engineering

Experience

Formula Slug (FSAE) — Santa Cruz, CA
Embedded Software Engineer, Electrical Branch — *October 2025 – Present*

- Own **STM32 C/C++ firmware** for real-time vehicle data acquisition, from sensor input to CAN bus telemetry
- Scoped, built, and validated tray-temperature monitoring across 4 DS18B20 OneWire sensors, triggering **automatic shutdown on safety-threshold faults**
- Replaced a discontinuous piecewise torque-mapping function with a **continuously differentiable cubic-polynomial curve**, now the **default accelerator mapping** across the car's VCU firmware
- Redesigned the CAN protocol for a new battery and chip architecture, **adopted team-wide** for on-car telemetry

Association for Computing Machinery (ACM), UCSC — Santa Cruz, CA
Web Development Lead — *August 2026 – Present*

- Lead a **team of 4 developers** through a full redesign of the club's website, delegating work across content, navigation, responsive UI, and backend authentication
- Direct delivery of **5 features**: member login with session handling, a unified navigation bar, a mobile-responsive layout, a redesigned events page, and a refreshed Meet the Board section
- Run weekly meetings to assign tasks and keep releases on schedule, reporting progress to the executive board

Projects

Sammy's Palate (UCSC Dining Tracker) | App Store | Live Web Demo — June 2026 – Present

- **Tech Stack:** Next.js, React, TypeScript, Supabase, PostgreSQL, Python, Playwright, Swift, Capacitor
- Launched a full-stack web and iOS app tracking daily nutrition from UCSC dining menus, **live on the App Store** with **60+ users**
- Built a scheduled Python/Playwright pipeline scraping **14 campus dining menus**, with name-based fallback
- Designed the PostgreSQL schema and Supabase **Row-Level Security** policies that enforce per-user data isolation
- Implemented email/password **authentication** with deep-linked iOS sign-in and password-reset flows
- Automated iOS releases with a Codemagic **CI/CD pipeline** that builds, code-signs, and ships each update

Heirloom (AI Multi-Generational Wealth Platform) — October 2026

- **Tech Stack:** Next.js, FastAPI, PostgreSQL, Neo4j, Clerk, Plaid, Bright Data, Daytona
- Team of 3 won **1st place in the Daytona track** (20 teams, 12 hours) at SF Tech Week, iterating on feedback from finance professionals
- Designed **AI agents** flagging family goal conflicts for wealth advisors, with **multi-tenant isolation** via RLS

Scaled Autonomous Vehicle (UC San Diego COSMOS) — July 2024 – August 2024

- **Tech Stack:** Python, OpenCV, ROS, LiDAR, GNSS, Jetson Nano
- On a **5-person team**, wrote Python drivers for a LiDAR and GNSS receiver and fused both into one time-aligned log, enabling **3 fully autonomous laps**
- Built **occupancy-grid mapping and localization**, validated in simulation and on real indoor/outdoor courses
- Generated **synthetic LiDAR logs** from a kinematic bicycle model to test mapping before hardware was ready